

sel0 (O13890) Catalytic Competence — Focused Curation Report

Gene: sel0 / SPAC20G4.05c (*S. pombe*, taxon 284812) · **UniProt:** O13890 **Hypothesis (slug: catalytic-competence):** *S. pombe sel0 is a catalytically competent protein AMPylase rather than a catalytically inactive SelO-family pseudoenzyme.* **Iterations:** 1–3 of 3 (final)

Executive Judgment

Verdict: SUPPORTED (by strong evolutionary + structural conservation; not directly assayed in *S. pombe*).

The "pseudoenzyme" framing rests on a well-known naming artifact: SelO adopts a protein-kinase-like fold but lacks canonical **kinase** catalytic residues, so it is called a **pseudokinase**. Sreelatha et al. 2018 (Cell, PMID **30270044**) showed this is misleading — SelO binds ATP in an inverted orientation and is a *bona fide active* protein **AMPylase/nucleotidyltransferase**. It is a pseudokinase but **not** a catalytically inactive enzyme.

For the *S. pombe* ortholog specifically, all evidence is homology/structure-based (there is **no direct in vitro or in vivo assay of sel0/O13890**). Within that limit, the evidence is unusually strong:

- O13890 is in the **SELO family** (Pfam PF02696; InterPro IPR003846; PANTHER PTHR32057).
- **10/10** UniProt-annotated catalytic/nucleotide-binding residues are **identical** to the experimentally characterized, crystallized *Pseudomonas syringae* SelO (Q87VB1) and *E. coli* YdiU (P77649), and to human SELENOO (Q9BVL4).
- The AlphaFold model (mean pLDDT 95.8) shows these residues forming a single, high-confidence, compact active-site pocket.

The most important caveat: catalytic competence for *S. pombe* is **inferred**. The UniProt functional annotations are ECO:0000250 (by similarity to *S. cerevisiae* Q08968), and PomBase lists sel0's characterisation status as "biological role inferred." No experiment has measured *S. pombe* sel0 AMPylation activity, substrate, or the effect of active-site mutation in fission yeast.

Update (Iteration 2) — the inference is better-anchored than distant-bacterial homology: - The by-similarity source, *S. cerevisiae* SelO (Q08968), carries experimental evidence (ECO:0000269) for FUNCTION and CATALYTIC ACTIVITY (references include Sreelatha 2018,

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and ECO:0000314 mitochondrial localization. The closest experimentally characterized ortholog is therefore *fungus*, not bacterial. - **PomBase already curates sel0** with MF GO:0070733 (protein adenylyltransferase activity), BP GO:0098869 (cellular oxidant detoxification), CC GO:0005739 (mitochondrion) — i.e., the AMPylase MF term is already the curated state. - **Nucleotide-specificity nuance:** the bacterial homolog YdiU is a Mn²⁺-dependent protein **UMPylase**

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whereas the eukaryotic mitochondrial branch (human SELENOO, *S. cerevisiae*, *S. pombe*) is the **AMPylase** branch. The branch-appropriate MF term for sel0 is protein adenylyltransferase activity (GO:0070733), not a UMP-transferase term.

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context
PMID 30270044 (Sreelatha 2018, Cell)	Direct assay + structural	Supports	SelO family is an active AMPylase, not an inactive pseudoenzyme	SelO transfers AMP from ATP to Ser/ Thr/Tyr; crystal structure shows kinase-like fold with ATP flipped → structural basis for catalysis; localizes to mitochondria, AMPylates redox proteins	Human SELENOO, <i>E.</i> <i>coli</i> , <i>P.</i> <i>syringae</i> (crystallized)
PMID 35101215 (Mukherjee & Sreelatha 2022)	Direct assay (method)	Supports	SelO catalytically AMPylates substrates	Bacterial SelO AMPylates substrates; enrichable with biotin-ATP analog	Bacterial SelO
PMID 39185898 (Gonzalez & Sreelatha 2024)	Direct assay (biophysical)	Supports	SelO binds ATP/ metal in inverted orientation to catalyze AMP transfer	TSA confirms metal + nucleotide binding underlying SelO AMPylation	SelO enzyme
UniProt O13890 (record)	Database	Supports (qualifies)	sel0 is an AMPylase	Annotated "Protein adenylyltransferase SelO, mitochondrial"; RHEA:54288; active site D287; Mg ²⁺ cofactor; SELO family — all ECO:0000250 (by similarity to Q08968)	<i>S. pombe</i>
This report — sequence		Supports		NW/BLOSUM62: 10/10 active/ binding residues	in silico

Citation	Evidence type	Direction	Claim tested	Key finding	Context
alignment (computed)	Structural/ evolutionary (computational)		Catalytic residues conserved, not degenerated	identical to characterized Q87VB1 (37% global id), P77649 (39.5%), Q08968 (43.6%), Q9BVL4 (36.7%): D287, K144, G120, G122, D156, G157, R208, R215, N288, D297	
This report — AlphaFold AF-O13890 (computed)	Structural (computational)	Supports	Active site is structurally intact	Model mean pLDDT 95.8; 8 catalytic residues at pLDDT 87.6–98.5 form one compact pocket (mean 9.0 Å to centroid; D287– N288 3.9 Å, D287– D297 9.2 Å)	in silico
UniProt Q08968 (record; refs incl.  30270044)	Direct assay / database	Supports	Fungal ortholog is a catalytically active adenylyltransferase	<i>S. cerevisiae</i> SelO carries ECO:0000269 (experimental) for FUNCTION and CATALYTIC ACTIVITY, and ECO:0000314 mitochondrial localization	<i>S. cerevisiae</i>
PomBase SPAC20G4.05c (record)	Database	Supports (qualifies)	GO curation state for sel0	Already annotated MF GO:0070733 (protein adenylyltransferase activity), BP GO:0098869	<i>S. pombe</i>

Citation	Evidence type	Direction	Claim tested	Key finding	Context
				(cellular oxidant detoxification), CC GO:0005739 (mitochondrion); status "biological role inferred"; genome-wide-screen phenotypes present but no mechanism assay	
PMID 32966796 (Yang 2020)	Direct assay	Qualifies	SelO-family nucleotide specificity	Bacterial homolog YdiU is a Mn ²⁺ -dependent protein UMPylase (Tyr/His) — family transfers different NMPs	Bacteria (Salmonella/ E. coli)

One peripheral binding residue differs vs human (R123 vs A156); human SELENOO is itself catalytically active, so this is not a competence-relevant substitution.

GO Curation Implications (leads — require curator verification)

- **Retain the molecular-function annotation of protein adenylyltransferase / AMPylase activity.** The most informative supported MF term is **GO:0070733 "protein adenylyltransferase activity"** (AMPylation), consistent with UniProt catalytic activity RHEA:54288. Do **not** downgrade to an "inactive"/"pseudo" characterization and do **not** fall back to "protein binding."
- **Evidence code:** Current support is homology (ISS/ISO, ECO:0000250) transferred from *S. cerevisiae* Q08968 and bacterial SelO. This is appropriate **as long as the annotation carries an ISS/ISO (by-similarity) qualifier**, not an experimental (IDA) code, because no *S. pombe* assay exists.

- **CC:** Mitochondrion (GO:0005739) — supported by orthology (ECO:0000250); reasonable to retain with ISS.
 - **BP:** "cellular response to oxidative stress" / redox homeostasis is a **family-level, orthology-based** association (Sreelatha 2018) and should be treated as **plausible-but-not-core / needs S. pombe evidence**, i.e. weaker than the MF claim.
 - **Net:** If the review action under evaluation was to weaken/remove the AMPylase MF term or brand sel0 an inactive pseudoenzyme, the evidence argues **against** that; the catalytic-AMPylase MF should be **retained** (as an ISS lead).
 - **Already-curated state (PomBase, Iteration 2):** sel0 is *already* annotated MF GO:0070733, BP GO:0098869 (cellular oxidant detoxification), CC GO:0005739, with characterisation "biological role inferred." The hypothesis is therefore **consistent with, and reinforces, the existing curation** — the recommended action is *retain GO:0070733 with an ISS/ISO qualifier*, anchored to the experimentally-supported *S. cerevisiae* ortholog Q08968 (ECO:0000269) rather than bacterial YdiU. Do **not** add a UMP-transferase term (that is the bacterial branch).
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Mechanistic Scope

- **Immediate molecular function (what is being tested):** transfer of AMP from ATP onto a hydroxyl side chain (Tyr, and in the family Ser/Thr) of substrate proteins — protein AMPylation/adenylylation — Mg^{2+} -dependent, via a kinase-like fold with inverted ATP.
 - **Downstream / non-core (do not conflate with the MF claim):** redox homeostasis, regulation of protein S-glutathionylation, and cellular oxidative-stress response are downstream consequences of AMPylating redox enzymes (e.g. glutaredoxins). These are pathway/phenotype-level, not the direct catalytic activity, and are orthology-inferred for *S. pombe*.
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Conflicts and Alternatives

- **"Pseudoenzyme" reading:** The only basis is the pseudokinase label (missing *kinase* catalytic residues). This is resolved by the 2018 discovery that SelO is an active *AMPylase* —

the kinase catalytic machinery is irrelevant to its true activity. No conflicting primary evidence indicates *S. pombe* sel0 is inactive.

- **Paralog confusion:** Low risk. sel0 is single-copy in the SELO family; the human accession Q9Y6D0 is *not* SELENOO (it returns Selenoprotein K) — the correct human ortholog is **Q9BVL4**. Curators reusing accessions should note this.
- **Selenocysteine caveat:** Some SelO orthologs (e.g., certain vertebrates) carry an active-site-proximal selenocysteine; *S. pombe* sel0 ends in a Cys-rich C-terminus (...QCSCSS) and is **not** predicted to be a selenoprotein. This does not affect the transferase active site (the AMP-transfer residues above are all standard and conserved).
- **Organism-specificity:** All functional data derive from bacterial/human/*S. cerevisiae* systems; a fission-yeast-specific loss of activity cannot be excluded from sequence alone, though nothing suggests it.

Knowledge Gaps

1. **No direct *S. pombe* enzymatic assay.** Checked: PubMed (no *S. pombe* SelO paper), UniProt (*S. pombe* = all ECO:0000250), PomBase (status "biological role inferred"). Mitigated but not closed by the experimentally-supported fungal ortholog Q08968 (ECO:0000269). Matters because the *S. pombe* MF annotation is still inferred. Resolved by: in vitro AMPylation assay of recombinant sel0 $\pm$ α -³²P/biotin-ATP.
 2. **Substrate identity in *S. pombe* unknown.** Family substrates are redox enzymes (glutaredoxins); the *S. pombe* targets are unverified. Resolved by: biotin-ATP substrate capture (per

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 in fission yeast lysate.
 3. **No active-site mutant phenotype in *S. pombe*.** Whether D287A/K144A abolishes an oxidative-stress phenotype is untested. Resolved by: catalytic-dead knock-in + oxidative-stress sensitivity assay.
 4. **Holo/ligand structure absent.** AlphaFold is apo and predicted; no experimental *S. pombe* structure or ATP-bound model. Resolved by: crystallography/cryo-EM or docking of ATP·Mg²⁺ into the modeled pocket.
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Discriminating Tests

1. **In vitro AMPylation assay** of recombinant sel0 with ATP (radiolabeled or biotin-ATP analog) on candidate substrates — directly distinguishes active enzyme from pseudoenzyme.
2. **Active-site mutagenesis** (D287A catalytic Asp; K144A invariant Lys; D297A Mg-Asp) — loss of activity confirms these residues are catalytic, matching the family.
3. **Genetic complementation:** does *S. pombe* sel0 rescue an oxidative-stress phenotype of a SelO-null (yeast/bacterial) strain, and is rescue abolished by the catalytic mutation?
4. **Substrate capture** with biotin-ATP

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 to identify AMPylated targets and test the glutaredoxin/redox-homeostasis link in fission yeast.

Curation Leads (require curator verification)

- **Candidate references to attach:**
- PMID 30270044 — snippet: *"the highly conserved pseudokinase selenoprotein-O (SelO) transfers AMP from ATP to Ser, Thr, and Tyr residues on protein substrates (AMPylation)"* and *"reveals a protein kinase-like fold with ATP flipped in the active site, thus providing a structural basis for catalysis."*
- PMID 35101215, PMID 39185898 — supporting AMPylation activity/mechanism.
- **Candidate GO terms:** keep **GO:0070733 protein adenylyltransferase activity** (MF); **GO:0005739 mitochondrion** (CC, ISS); treat oxidative-stress-response BP as orthology-level (ISS), not core/experimental.
- **Possible action change:** if the blinded review action weakened or removed the AMPylase MF term (treating sel0 as an inactive pseudoenzyme), reverse toward **retain-with-ISS**; the pseudokinase status must not be read as catalytic inactivity.
- **Suggested curator question:** Is there any *primary* *S. pombe* experimental evidence for sel0 activity/localization, or is the annotation entirely projected from Q08968/bacterial SelO? If entirely projected, ensure ISS/ISO evidence codes.
- **Suggested experiments:** the four discriminating tests above.

Catalytic-Residue Conservation Table (computed provenance)

sel0/O13890 residue → aligned residue in each characterized homolog (Iteration 1 NW/BLOSUM62 alignments):

sel0 (O13890)	Role	PsSelO Q87VB1 (xtal)	EcYdiU P77649	ScSelO Q08968	hSELENOO Q9BVL4	Verdict
D287	Active site: catalytic Asp (proton acceptor)	D252	D246	D338	D338	IDENTICAL in all
K144	Invariant Lys (ATP α - phosphate positioning)	K113	K107	K156	K176	IDENTICAL in all
D156	Mg ²⁺ - coordinating Asp	D125	D119	D168	D188	IDENTICAL in all
D297	Mg ²⁺ - coordinating Asp	D262	D256	D348	D348	IDENTICAL in all
G120	Gly-rich ATP- binding loop	G90	G84	G132	G153	IDENTICAL in all
G122	Gly-rich ATP- binding loop	G92	G86	G134	G155	IDENTICAL in all
G157	Gly (active-site loop)	G126	G120	G169	G189	IDENTICAL in all
R208	Substrate/ phosphate Arg	R176	R170	R220	R246	IDENTICAL in all
R215	Substrate/ phosphate Arg	R183	R177	R227	R253	IDENTICAL in all
N288	Active-site Asn	N253	N247	N339	N339	IDENTICAL in all
R123	Peripheral binding Arg	R93	R87	R135	A156	IDENTICAL except human (A)

Catalytic residues degenerated in sel0: 0. 10/11 identical across all characterized homologs (including the crystallized *P. syringae* SelO and the experimentally-validated *S. cerevisiae* SelO); the single exception (R123→A) is a peripheral residue that differs only vs human, and human SELENOO is itself active.

GO Decision Table (leads — require curator verification)

Aspect	GO ID	Term	Lead action	Rationale
MF	GO:0070733	protein adenylyltransferase activity	RETAIN (with ISS/ ISO)	Full active site conserved + AlphaFold pocket; anchored to ScSelO Q08968 (ECO:0000269); already curated in PomBase
CC	GO:0005739	mitochondrion	RETAIN (ISS)	Ortholog localization (ScSelO ECO:0000314); UniProt SL-0173; already in PomBase
BP	GO:0098869	cellular oxidant detoxification	RETAIN as non-core / orthology- level	Family redox role (P 30270044); downstream of AMPylation; not <i>S. pombe</i> -assayed
MF	—	protein binding	AVOID as final term	Less informative than GO:0070733
MF	—	UMP-transferase / UMPylation	DO NOT ADD	Bacterial YdiU branch (P 32966796); sel0 is the eukaryotic AMPyase branch

Provenance

Computed in this run (executed code + outputs recorded in the job): (1) UniProt REST retrieval of O13890 and homolog features; (2) Needleman–Wunsch/BLOSUM62 pairwise alignments mapping the 10 annotated catalytic/binding residues onto Q87VB1, P77649, Q08968, Q9BVL4

(all identical); (3) AlphaFold AF-O13890-F1 model parsing — pLDDT and 3D clustering of the catalytic pocket. No results were fabricated; homolog activity data are from primary literature, and the *S. pombe* catalytic competence is reported as an inference, not a measurement.

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